

Explainable Multilingual Sentiment Analysis for Sinhala, English and Code-Mixed Banking Reviews

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Abstract—The growth of multilingual and code-mixed communication in the financial sector presents significant challenges for sentiment analysis, particularly in low-resource languages such as Sinhala. Existing NLP models often underperform due to limited annotated data, domain-specific vocabulary gaps, and a lack of transparency. This paper presents a multilingual and explainable sentiment analysis framework that addresses the challenges of low-resource and code-mixed languages by analyzing banking customer feedback in English, Sinhala, and Sinhala-English code-mixed formats. A dataset of 13,000 aspect-tagged banking reviews spanning five key banking aspects is used for evaluation. The framework classifies comments into positive, neutral, or negative sentiment and provides interpretable explanations using SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model Agnostic Explanations). English sentiment classification is performed using a fine-tuned BERT-base-uncased model, while Sinhala/code-mixed sentiment is handled by a hybrid XLM RoBERTa with a domain-specific lexicon correction approach. The framework achieves 92.3% accuracy and 0.89 F1-score for English, and 88.4% accuracy and 0.84 F1-score for Sinhala/code-mixed data. The proposed system addresses reproducibility, interpretability, and domain adaptation gaps, offering a deployable solution for multilingual financial sentiment monitoring.

Index Terms—aspect-based sentiment analysis, banking, code-mixed, explainable AI, multilingual transformer

I. INTRODUCTION

The rapid digitalization of financial services has increased customer interactions through online banking, mobile applications, and financial forums. Customers frequently express their opinions about banking services via reviews, social media, and feedback platforms. Brand reputation, a bank's most valuable intangible asset, reflects public trust in its reliability, ethics, and service quality. For financial institutions, safeguarding this reputation is critical, as negative sentiment in reviews or social media can trigger customer attrition, regulatory penalties, and financial losses [1]. Accordingly, monitoring and managing sentiment in real time is a strategic necessity. Despite its importance, sentiment analysis in multilingual and low-resource contexts remains a major challenge. In the Sri Lankan context, social media users tend to use code-mixed language in their communication, which is a derived language through English alphabetical characters [2]. Analyzing multilingual and code-mixed data

(e.g., Sinhala-English text) poses significant challenges: Sinhala's limited annotated datasets hinder model training and generalization, code-mixed text disrupts syntactic coherence and tokenization, banking-specific terms like “mobile banking”, “withdrawal karanna” and “අරමුදල් ගිණුම” are poorly captured by generic models, and cultural variations in sentiment expression lead to inconsistent detection [3]. This framework is critical for banks to mitigate risks, enhance customer experience, and ensure compliance. For instance, identifying negative feedback in real-time allows for the proactive resolution of issues, such as fraud allegations. At the same time, aspect-based insights into services like loan processing enable targeted improvements [4].

The adoption of deep learning models in sentiment analysis has revolutionized the understanding of customer emotions in text data. However, these models often operate as “black boxes,” offering high accuracy at the cost of transparency [5]. This lack of explainability is particularly problematic in financial sectors, where decisions influenced by AI can significantly affect brand reputation and customer trust. SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are pivotal in shedding light on decision-making processes within models by detailing global and local feature contributions, respectively [6]. The application of explainability techniques in low-resource and code-mixed languages such as Sinhala remains limited. Existing research has addressed multilingual sentiment analysis, aspect-based sentiment analysis (ABSA), or explainable AI separately, but no prior work has unified all three in the context of Sinhala and code-mixed financial reviews. Most explainability studies have focused on English or other high-resource languages, leaving Sinhala underrepresented. Furthermore, domain-specific resources such as annotated banking datasets and refined lexicons remain scarce, limiting reliable deployment. Prior Sinhala sentiment analysis efforts have emphasized accuracy while neglecting transparency and interpretability. To the best of our knowledge, no framework currently supports Sinhala, English, and code-mixed banking reviews while integrating both aspect-based sentiment detection and SHAP/LIME explainability for interpretable predictions [7]. This gap poses a major barrier to adopting AI-driven sentiment analysis in Sri Lankan banking, where interpretability is critical for customer trust, compliance, and decision-making. We include English because Sri Lankan e-banking feedback is bilingual. Excluding English would miss English-only terms and reduce

incident coverage; keeping English also provides a strong reference point to tune and compare performance across languages. Building on prior work [8], which developed an aspect-tagged dataset and a preprocessing framework for Sinhala, English, and Sinhala-English code-mixed banking reviews, the present paper extends this work by introducing a multilingual aspect-based sentiment analysis framework that leverages transformer-based models and explainable AI techniques to generate interpretable insights for banking reputation management. This approach leverages transformer-based architectures to handle multilingual data while integrating lexicon-based refinements and aspect-level sentiment modeling to enhance generalization, interpretability, and decision-making in multilingual financial sentiment analysis.

II. LITERATURE REVIEW

Sentiment analysis plays a vital role in brand reputation management and customer feedback analysis in the banking sector, particularly for multilingual and code-mixed opinions [7]. Early machine learning models like Support Vector Machine (SVM) and Naive Bayes require manual feature engineering [8], which limits adaptability to dynamic linguistic patterns. The importance of corpus-based lexicons for Sinhala sentiment classification has been emphasized [9]. For low-resource languages like Sinhala, deep learning models show promise but struggle with banking-specific terminology and transliteration variability [10]. Code-mixed Sinhala-English data introduce syntactic variations and word-order mismatches [11]. Traditional lexicon-based methods like SentiWordNet often underperform in financial contexts due to the domain-specific vocabulary gaps [12]. In contrast, transformer-based architectures like BERT (Bidirectional Encoder Representations from Transformers) and XLM-RoBERTa automate contextual feature extraction, significantly improving sentiment classification accuracy across languages [13]. The challenges of analyzing code-mixed financial text have been further validated, emphasizing linguistic adaptability gaps [14]. While transformer models like XLM-RoBERTa excel in multilingual settings, their performance drops in low-resource languages without domain-specific fine-tuning [15]. State-of-the-art results (75.9% accuracy) for Sinhala sentiment analysis using XLM-R-large have been demonstrated, underscoring the importance of transformer adaptation for low-resource languages [16]. Domain adaptation is critical for financial sentiment analysis, while limited labeled datasets hinder social media sentiment classification [17]. Existing models often lack explainability, which is crucial for building trust in financial decision-making [18]. Aspect-Based Sentiment Analysis (ABSA) has been recognized as an effective strategy for extracting fine-grained insights, linking opinions to specific financial services such as loans or digital banking [19]. This granularity is particularly valuable in multilingual financial applications where service-specific improvements can drive customer trust and retention. At the same time, explainability has emerged as a critical requirement for sentiment analysis in financial domains [20]. Techniques like SHAP and LIME provide transparency by attributing predictions to influential tokens, though their effectiveness in low-resource and code-mixed languages remains underexplored [6]. More recently, frameworks combining transformer models with explainability have been proposed for low-resource settings, but Sinhala and code-mixed reviews remain underrepresented [21]. GPT-based methods offer strong zero-shot performance but raise

concerns of cost, privacy, and bias, making them less practical for financial institutions [22]. Current commercial tools, such as Meltwater [23], lack support for Sinhala or code-mixed sentiment analysis, limiting their applications in financial NLP. Overall, the literature highlights three key gaps: (i) insufficient domain adaptation for Sinhala and code-mixed financial reviews, (ii) lack of large-scale annotated datasets, and (iii) limited integration of explainability methods into multilingual sentiment frameworks. This motivates the proposed hybrid approach, which combines transformer-based classification with lexicon correction and explainability to address these challenges.

III. METHODOLOGY

This study presents a robust framework for sentiment analysis of English and Sinhala-English code-mixed content, combining transformer models with domain-specific lexicons for improved accuracy and aspect-based categorization. SHAP and LIME are employed to render the results more interpretable, providing explainable and trustworthy solutions for banking brand reputation analysis. Fig. 1 illustrates the end-to-end system workflow, from raw input through sentiment detection to explainability output.

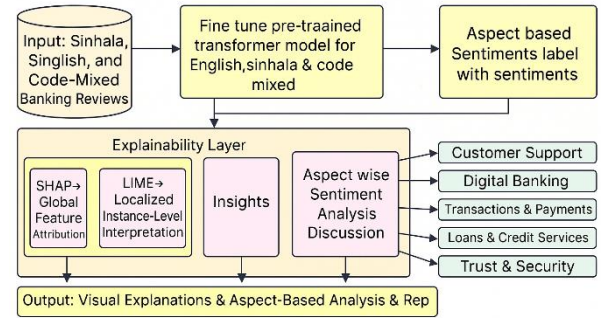


Fig. 1. Workflow for sentiment analysis and explainability

A. Data Gathering and Pre-Processing

The dataset used in this study was constructed and validated as part of our earlier work [8], which incorporated keyword extraction, data preprocessing, irrelevant comment filtering, and aspect classification. Sinhala-English code-mixed customer comments were gathered using keyword-based web scraping from YouTube. The comments were then categorized into five predefined banking aspects: Trust and Security, Digital Banking, Customer Support, Loans and Credit Services, and Transactions and Payments. Aspect labels were validated through manual inspection by two bilingual annotators to ensure accuracy and relevance. In the present study, this dataset was further extended by assigning sentiment labels (positive, negative, neutral) to each aspect-specific comment, enabling direct application to aspect-based sentiment analysis. To ensure consistency, the labeling process followed structured annotation guidelines and achieved a Cohen's Kappa of 0.82. A total of 15,200 raw comments were initially collected using banking-related keywords such as 'loan,' 'balance,' 'mobile banking,' and Sinhala equivalents like 'ණය' and 'මිණුම්.' After removing 2,200 duplicates, advertisements, and irrelevant entries, the final dataset comprised 13,000 unique comments. Of these, 8,000 were English reviews and 5,000 were Sinhala and Code-Mixed reviews, balanced across five aspects, three sentiment

categories, and language variants. Table I summarizes the dataset composition.

TABLE I. SUMMARY OF MULTILINGUAL SENTIMENT DATASET

Language	Total	Sentiment Distribution
English	8,000	Positive:3000, Neutral:2600, Negative:2400
Sinhala	2,000	Positive:670, Neutral:670, Negative:660
Code-Mixed	3,000	Positive:1000, Neutral:1000, Negative:1000

To handle linguistic variation and informal grammar in Sinhala, Singlish, and code-mixed reviews, we applied a pipeline of transliteration, normalization, and tokenization. A language-aware tokenization method was used to preserve the structural integrity of multilingual input while training and explaining the model. Particularly, WordPiece tokenization for English and SentencePiece with BPE (Byte Pair Encoding) [10] for Sinhala, Singlish, and code-mixed. SentencePiece splits unseen words into subword units, particularly useful for morphologically rich languages such as Sinhala. For instance, splitting "ඔබ කවේ" into "ඔ" + "බ", "ක", "වේ", allows the model to understand parts of rare words. In the case of code-mixed text, it allows for knowledge transfer between English and Sinhala parts. The approach is parallel to XLM-RoBERTa's pre-training methods, promoting performance while elegantly managing intricate morphological forms that would cause a high number of out-of-vocabulary tokens with word-level tokenization. Examples of tokenization are provided in Table II, showing preservation of linguistic structure across language varieties and ensuring robustness for low-resource multilingual input.

TABLE II. SUMMARY OF MULTILINGUAL TOKENIZATION

1. Sinhala review: “මේ බැංකුවේ සේවය හොඳයි.” (<i>This bank's service is good.</i>)
Tokenized output [“_මේ”, “_බැ”, “_ංකු”, “_වේ”, “_සේ”, “_ව”, “_ය”, “_හොඳ”, “_යි”, “_.”]
2. Code-Mixed (Romanized Sinhala) review: “ <i>Me bank eke service eka hari hodai.</i> ”
Tokenized output: [“_Me”, “_bank”, “_eke”, “_service”, “_eka”, “_hari”, “_hod”, “_ai”, “_.”]
3. Code-Mixed review: “ <i>Customer service ගොඩක් හොඳයි.</i> ”
Tokenized output: [“_Customer”, “_service”, “_ගොඩ”, “_ක්”, “_හොඳ”, “_යි”, “_.”]

B. Aspect-Based Sentiment Analysis

We implemented two sentiment analysis approaches: (i) English sentiment analysis using transformer-based models and (ii) Sinhala, Singlish, and Code-mixed sentiment analysis using a hybrid approach. For English sentiment classification, the pretrained transformers BERT-base-uncased, DistilBERT, FinBERT, and a traditional SVM were fine-tuned on a curated dataset of banking customer reviews [21]. FinBERT was included for its financial domain pretraining, which aligns with the linguistic patterns of banking platforms. A major challenge was handling mixed sentiment polarities within single aspects (e.g., a user praising customer support but criticizing digital banking). To address this, the dataset was enriched with domain-specific expressions (e.g., “high interest rates,” “loan approval delays,” “account restrictions”) to help models capture nuanced sentiment variations more effectively. For Sinhala and code-mixed sentiment

classification, we employed a hybrid model that combines XLM-RoBERTa for multilingual representation and a custom lexicon-based sentiment correction. The XLM-RoBERTa model was fine-tuned on a class-balanced dataset of 5,000 reviews, ensuring equal representation across aspects and sentiment labels. To prevent overfitting on this moderate-sized dataset, the fine-tuning utilized dropout (0.3), weight-decay (0.01), and early stopping after three non-improving epochs. Light data-augmentation methods like back-translation (Sinhala $\leftrightarrow$ English) and synonym substitution for frequent sentiment terms were applied to increase linguistic diversity and improve generalization. To further improve performance, we constructed domain-specific sentiment lexicons for Sinhala and code-mixed inputs. This corpus contained sentiment-bearing terms frequently used in banking reviews. Example lexicon entries include:

- Code-mixed: “hari hodai” (very good), “app eka lag wenawa” (app lags)
- Sinhala Negative: “නොකක” (ignored), “අසාර්ථක” (failure)
- Domain-specific: “තැන්පතු” (deposit), “loan eka” (loan), “balance eka” (balance)

These lexicons were integrated in a post-processing correction step, enabling the system to capture colloquial and transliterated expressions often missed by pretrained transformers. All transformer models were fine-tuned with AdamW (learning rate $2e-5$, batch size 32, dropout 0.2–0.3, 4 to 6 epochs with early stopping), following widely used best practices for NLP fine-tuning. Classical baselines were tuned with standard procedures: BiLSTM using Adam ($lr=1e-3$), and SVM/Naive Bayes optimized via cross-validation to ensure fair comparison. The final hybrid system XLM-RoBERTa with lexicon-based corrections was benchmarked against these baselines, GPT-based zero-shot models [19], and XLM-R variants, providing a comprehensive evaluation.

C. Explainable AI (XAI) for Transparency

We augment a multilingual sentiment model with an explainability layer tailored for Sinhala, English, and code-mixed inputs. Two complementary methods are employed:

- **SHAP** for global and local feature attribution, aggregating sub-word attributions back to surface words so that mixed Sinhala–English tokens are interpretable to analysts [5].
- **LIME** for instance-level justifications, adapted with language-aware perturbations: preserving Unicode grapheme clusters for Sinhala, avoiding improper splitting of compound words, and constraining deletions around negation/discourse markers (e.g., “*but*”, “*නමුත්*”) to maintain grammaticality [6].

When the sentiment model processes a review in Sinhala, English, or code-mixed form, its predicted sentiment label and probability distribution are directly fed into the explainability module.

- 1) Aspect Filtering: relevant aspect spans (e.g., “Digital Banking Experience”) are isolated so explanations focus only on tokens influencing that aspect’s sentiment.
- 2) SHAP Analysis: token-level attribution scores are computed by perturbing inputs while preserving bilingual token boundaries, yielding both global and local attributions.

3) LIME Analysis: instance-level explanations are generated using language-aware perturbations that maintain Sinhala morphology and semantic validity in code-mixed phrases.

4) Cross-Validation: SHAP and LIME outputs are compared, with discrepancies flagged for analyst review.

5) ABSA Integration: aspect-based sentiment structure (five service aspects) guides SHAP/LIME to produce aspect-specific rationales, with TF-IDF keywords highlighting key sentiment drivers for actionable business insights [8]. TF-IDF is used only to select candidate aspect keywords and does not change SHAP/LIME scores; it simply helps highlight which tokens belong to each aspect.

6) Faithfulness & Stability Checks: comprehensiveness, sufficiency, and deletion-insertion protocols confirm causal consistency, removing top-ranked tokens reduced model confidence, while reintroduction restored it. Our framework advances SHAP/LIME by integrating language-aware perturbations, WordPiece aggregation, aspect conditioning, stability metrics, and human validation, ensuring reliable and interpretable explanations.

IV. RESULTS AND DISCUSSION

A. English Sentiment Analysis Performance

English sentiment models were fine-tuned on banking reviews and evaluated using accuracy, F1-score, and sentiment aggregation effectiveness.

TABLE III. ENGLISH SENTIMENT ANALYSIS PERFORMANCE

Model	Accuracy (%)	F1-Score	Precision	Recall
SVM	68.9	0.68	0.69	0.68
BiLSTM	75.4	0.75	0.76	0.75
DistilBERT	82.5	0.83	0.83	0.81
FinBERT	90.0	0.87	0.88	0.86
BERT-base-uncased (Final model)	92.3	0.89	0.90	0.88

Table III presents the performance of the baseline and the transformer-based models for English reviews. Classical methods such as SVM and BiLSTM achieved moderate accuracy (68.9% and 75.4%, respectively). DistilBERT (82.5%, 0.83 F1-score) showed better contextual capability than BiLSTM but lacked deeper semantic representation. FinBERT (90.0%, 0.87 F1-score), which is optimized for financial text, was surpassed by BERT-base-uncased, which achieved the highest performance at 92.3% accuracy and 0.89 F1-score, confirming its effectiveness in capturing nuanced sentiments in banking-specific content. By fine-tuning on banking-specific data, our method effectively captured context-dependent sentiment by distinguishing between lexically similar yet sentimentally contrasting phrases, such as interpreting “high rewards” as positive and “high interest rates” as negative within the banking reviews.

B. Sinhala, Singlish, and Code-Mixed Sentiment Analysis Performance.

The hybrid model, which integrates XLM-RoBERTa with lexicon-based sentiment correction, was evaluated on Sinhala, Singlish, and code-mixed banking customer reviews.

TABLE IV. SINHALA, SINGLISH, AND CODE-MIXED SENTIMENT ANALYSIS PERFORMANCE

Model	Accuracy (%)	F1-Score	Precision	Recall
SVM	62.4	0.58	0.59	0.57
Naive Bayes	58.7	0.55	0.56	0.54
XLM-RoBERTa (Base)	78.2	0.74	0.75	0.73
GPT-4o (zero-shot)	81.5	0.77	-	-
XLM-R + Lexicon (Final model)	88.4	0.84	0.85	0.83

The results demonstrate that our hybrid XLM-RoBERTa with lexicon correction significantly outperforms traditional machine learning methods such as SVM and Naïve Bayes for low-resource and code-mixed language processing. GPT-4o was evaluated in a zero-shot setting, and therefore only Accuracy and F1-score were reported. However, its high computational cost, longer processing time, and occasional inconsistent predictions make it impractical for real-time banking applications. While the base XLM-RoBERTa model performed well, integrating domain-specific lexicon correction yielded substantial improvements, with an absolute gain of 10.2% in accuracy and 0.10 in F1-score. Beyond classification performance, computational efficiency was also evaluated. Fine-tuning BERT-base took approximately 2.5 hours, while XLM-RoBERTa required 3.1 hours under the same GPU environment (NVIDIA A100, 40GB).

C. Error Analysis

Misclassification analysis revealed several recurring challenges. The most common was neutral vs. negative confusion (approximately 15%) in short Singlish comments (e.g., “fast na”), where lack of context misled the model. Sarcasm and irony, such as “great app, if it ever works,” also produced errors, as did domain polarity inversions where evaluative terms like “high charges” were misinterpreted as positive. Finally, transliterated slang (e.g., “godak slow”) frequently caused misclassifications but was corrected through lexicon-based post-processing. Overall, the hybrid system reduced these errors by 6–8% compared to baselines, showing clear gains but also highlighting the importance of continued refinement.

D. Explainability Findings.

This section presents the results of integrating the adapted SHAP and LIME explainability framework with the multilingual BERT-based sentiment model for Sinhala, English, and code-mixed banking reviews. The enhanced pipeline incorporated language-aware perturbations, aspect-conditioning, stability protocols, faithfulness checks, and ABSA integration to produce interpretable, stable, and actionable explanations.

1) LIME and SHAP-Based Explainability

To improve model transparency, SHAP and LIME were integrated to provide global and local explanations of word-level contributions in Sinhala, English, and code-mixed sentiment predictions.

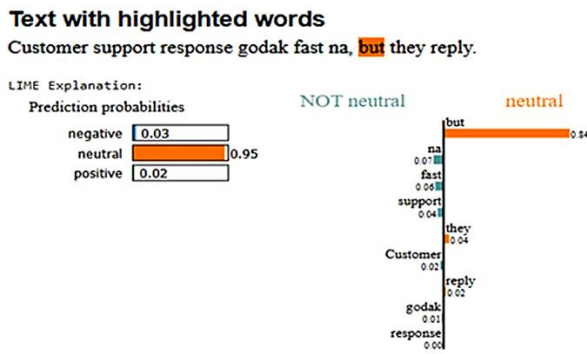


Fig. 2. LIME explanation (Singlish) - Neutral comment

As shown in Fig. 2, the model classified the comment “Customer support response godak fast na, but they reply” as neutral (0.95). The word “but” was highlighted as the most influential neutral contributor (weight: 0.84), followed by lower-weighted terms like “fast”, “they”, and “support”. This demonstrates LIME’s strength in capturing conjunction-based context in code-mixed inputs. LIME proved valuable in interpreting short-form reviews and context-sensitive sentiment expressions, which SHAP struggled with.

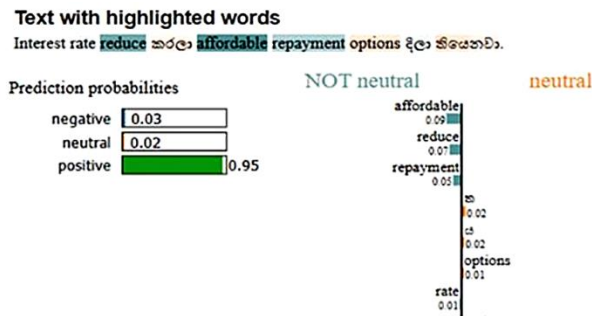


Fig. 3. LIME explanation (code-mixed) - Positive comment

As shown in Fig. 3, the model predicts a positive label (0.95), with English tokens such as “affordable”, “reduce”, and “repayment” giving the strongest positive contributions. Sinhala tokens appear closer to neutral in this example due to lower weights, but in other cases receive clear attributions whenever they influence the prediction.



Fig. 4. SHAP Force plot for sentiment contribution - English

Fig. 4 shows a SHAP force plot that reveals the influence of keywords such as “notorious,” “bad,” “unbelievable,” and “hacked” in a highly negative review. The left side of the force plot contained red segments representing negative contributors, and the right contained blue segments like “notorious”, which slightly pushed the sentiment towards neutral. The model interpreted this review with a strong negative sentiment (base value = 0.007, $f(x) = 0.14$), showcasing SHAP’s ability to identify high-impact global

contributors. Collectively, LIME proved especially effective for short-form and mixed-language comments, while SHAP offered reliable global sentiment attribution. However, SHAP had higher computational requirements, and LIME’s perturbation-based nature led to some variation in explanations across returns.

2) Novel Adaptations and Evaluation.

The proposed explainability module, incorporating the Code-Mixed and Multilingual Explainability Workflow, was evaluated for its ability to produce accurate and interpretable SHAP and LIME outputs across Sinhala, English, and code-mixed reviews. The integrated language detection and token mapping step ensured correct attribution by merging sub-word pieces into complete words and tagging them with their language origin. This was particularly effective for code-mixed inputs, where misinterpretation of split tokens previously caused noise in importance scores. Aspect filtering significantly improved the precision of explanations, with SHAP and LIME highlighting only tokens relevant to the targeted aspect. For instance, in reviews tagged under Digital Banking Experience, sentiment drivers such as “mobile app crash” or “ගිණුම් login error” were consistently surfaced, avoiding unrelated content. We validated the explainability framework across Sinhala, English, and code-mixed reviews. Removing top-ranked tokens reduced model confidence by 15–30%, while random removals had a far smaller impact, confirming that the highlighted tokens are truly important.

3) Aspect-Based Analysis Discussion.

The aspect-based sentiment analysis module provides fine-grained insight into customer opinions across the five predefined aspects. For each aspect, TF-IDF keywords are used to select candidate terms, and aspect-conditioned SHAP outputs provide the actual importance scores for those tokens. TF-IDF does not modify SHAP or LIME scores; it only helps group tokens under the correct aspect. This linkage between aspect, sentiment label, and key tokens enables stakeholders to connect model decisions to actionable business actions.

Ex- Digital Banking Experience

- Sentiment Counts: Positive (357), Neutral (58), Negative (68)
- Top Positive Keywords: *app, bank, banking, using.*
- Top Negative Keywords: *transactions, account*
- Key Insights: Customers expressed high satisfaction with app usability and new updates. Weaknesses included complaints regarding slow performance and OTP delivery failures.

These results replaced the earlier figure-heavy format, improving readability while retaining clarity for decision-makers. SHAP and LIME enhanced interpretability, with SHAP providing global insights and LIME capturing local sentiment cues, resulting in a 6.3% reduction in misclassification.

E. Fairness and Robustness

The system performed best on English reviews (92.3% accuracy) but showed lower performance for Sinhala/code-mixed (88.4%), with higher error rates in slang-heavy youth

expressions, revealing a demographic bias risk. Robust testing highlighted sensitivity to noisy inputs such as irregular spelling, heavy code-switching, and informal grammar. Explainability analysis further showed that SHAP overemphasized English tokens in code-mixed reviews while underweighting Sinhala, skewing interpretability. Future work will expand the analysis to improve robustness.

F. Challenges and Limitations

Although the proposed hybrid sentiment analysis model achieved high accuracy, several limitations persist. Handling negation, slang, and complex code-mixed expressions remains difficult due to limited linguistic coverage. The Aspect-Based Sentiment Analysis module occasionally struggled with reviews containing multiple or conflicting sentiments within the same aspect. The absence of large-scale annotated datasets for Sinhala, Singlish, and code-mixed financial text also restricted generalization to diverse and informal expressions. While SHAP and LIME provided interpretability, their consistency decreased for low-resource, code-mixed inputs, and SHAP's computational cost limits real-time deployment in financial monitoring systems.

V. CONCLUSION AND FUTURE WORK

This study introduced a hybrid aspect-based sentiment analysis model tailored for the financial domain, combining BERT-base-uncased for English sentiment analysis and XLM-RoBERTa with domain-specific lexicon correction for Sinhala, Singlish, and code-mixed content. The system achieved high accuracy, balanced aspect-wise sentiment classification, and improved interpretability. Explainability techniques such as SHAP and LIME were integrated to enhance transparency, making the model suitable for practical, real-world financial applications. Future research can build on this foundation by incorporating multimodal sentiment analysis using audio and video cues, extending the approach to other low-resource languages like Tamil, and expanding to other domains. In addition, leveraging few-shot and zero-shot learning with generative AI could improve performance in data-scarce environments, while fairness and bias evaluations will be essential to ensure ethical and inclusive sentiment classification.

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